Day 056 | Unit 04 | Teacher plan

### Numeric and Algebraic Expressions

# Generate Equivalent Expressions

Standards: 6.EE.A.3

Learning target: I can use distribution to generate an equivalent expression.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply the factor outside parentheses by every term inside. The two expressions have the same value for every allowed input. An area model can show how one rectangle splits into smaller rectangles.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 16

Day 056 | Unit 04 | Lesson

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## 01 / Learn

Multiply the factor outside parentheses by every term inside. The two expressions have the same value for every allowed input. An area model can show how one rectangle splits into smaller rectangles.

## Worked example 1

Expand 5(x + 3).

1. Multiply the factor outside parentheses by every term inside.

2. The two expressions have the same value for every allowed input.

3. Multiply the outside factor by every inside term, including the constant.

4. State the result in the form requested: 5x + 15.

Answer: 5x + 15

## Worked example 2

Select all expressions equivalent to 6(x + 6).
A. 6x + 6
B. 12x
C. 6x + 36
D. 36 + 6x

1. Multiply the factor outside parentheses by every term inside.

2. Distribute 6 to both terms to get 6x + 36.

3. Reversing the addends gives the same value.

4. The other expressions fail, for example at x = 0.

Answer: C, D - 6x + 36; 36 + 6x

# Day 056 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

Expand 2(x + 6).

### 02 / Constructed response

Expand 3(x + 6).

### 03 / Constructed response

Expand 3(x + 7).

### 04 / Constructed response

Expand 2(4x + 4).

### 05 / Constructed response

Expand 3(4x + 2).

### 06 / Constructed response

Expand 7(7x + 6).

## 07 / Constructed response

Expand 2(6x + 5). Verify your result at x = 2.

## 08 / Multiple select

Select all expressions equivalent to 3(x + 4).
A. 12 + 3x
B. 7x
C. 3x + 4
D. 3x + 12

## 09 / Multiple select

Select all expressions equivalent to 4(x + 3).
A. 12 + 4x
B. 7x
C. 4x + 3
D. 4x + 12

## 10 / Constructed response

Expand 5(7x + 2). Verify your result at x = 2.

Day 056 | Unit 04 | Answer key

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### 1. 2x + 12

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

### 2. 3x + 18

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

### 3. 3x + 21

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

### 4. 8x + 8

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

### 5. 12x + 6

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

### 6. 49x + 42

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

### 7. 12x + 10; both give 34 at x = 2

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

## 8. A, D - 12 + 3x; 3x + 12

Distribute 3 to both terms to get 3x + 12. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

## 9. A, D - 12 + 4x; 4x + 12

Distribute 4 to both terms to get 4x + 12. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

## 10. 35x + 10; both give 80 at x = 2

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response